- Part 1: Numpy and math Foundations

Apply element-wise operations

January_spending = [1200, 2500, 800, 4000]

February_spending = [1400, 2200, 1000, 4500]

1_Addition:

= [1200 + 1400, 2500 + 2200, 800 + 1000, 4000 + 4500]
= [2600, 4700, 1800, 8500]

2_Subtraction:

= [1200 - 1400, 2500 - 2200, 800 - 1000, 4000 - 4500]
= [-200, 300, -200, -500]

3_Multiplication:

= [1200 x 1400, 2500 x 2200, 800 x 1000, 4000 x 4500]
= [1680000, 5500000, 800000, 18000000]

4_Division:

= [1200 / 1400, 2500 / 2200, 800 / 1000, 4000 / 4500]
= [0,85, 1,13, 0,8, 0,88]

Apply aggregations

Spending = [1200, 2500, 800, 4000]

1_ Sum

= 1200 + 2500 + 800 + 400
= 8500

2_ Mean

= (1200 + 2500 + 800 + 400) / 4
= 2125

3_ Min

= 800 The minimum value

4_ max

= 4000 The maximum value

5_ Standard deviation

$(1200 - 2125)^2 = 855625$ $(800 - 2125)^2 = 1755625$
$(2500 - 2125)^2 = 140625$ $(4000 - 2125)^2 = 3515625$

$$\sigma^2 = \frac{855625 + 140625 + 1755625 + 3515625}{4} = 1556875$$

$$\sigma = \sqrt{1556875}$$

$$= 1247,75$$

Part 2 : Vectors and Linear Algebra

$v = [1200, 2500, 800, 4000]$

$w = [1400, 2200, 1000, 4500]$

1- vector addition

$$v + w = [1200 + 1400, 2500 + 2200, 800 + 1000, 4000 + 4500]$$
$$= [2600, 4700, 1800, 8500]$$

2- Scalar multiplication

$$2v = [2(1200), 2(2500), 2(800), 2(4000)]$$
$$= [2400, 5000, 1600, 8000]$$

3- Dot Product

$$v \cdot w = [(1200 \times 1400) + (2500 \times 2200) + (800 \times 1000) + (4000 \times 4500)]$$
$$= [1680000 + 5500000 + 800000 + 18000000]$$
$$= 25980000$$

4- Norm

$$\|v\| = \sqrt{(1200)^2 + (2500)^2 + (800)^2 + (4000)^2}$$
$$= \sqrt{24330000}$$
$$= 4932,54$$

5_ Cosine similarity

$$\text{CosineSimilarity}(v, w) = \frac{v \cdot w}{\|v\| \, \|w\|} = \frac{25980000}{4932.54 \times 5296,23} = 0,99$$

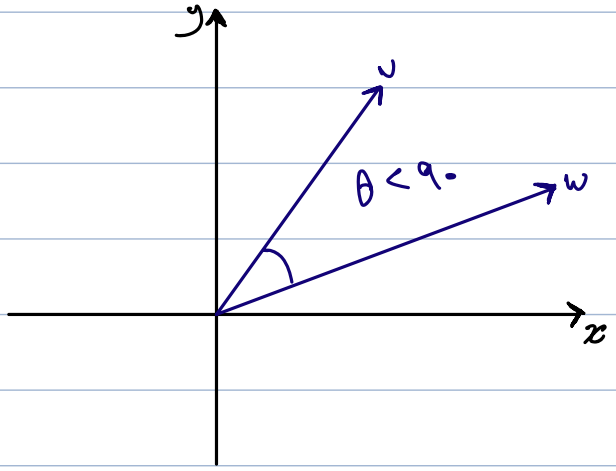
$$\|w\| = \sqrt{(1400)^2 + (2200)^2 + (1000)^2 + (4500)^2}$$

$$= 5296,23$$

- Explain the meaning of dot product

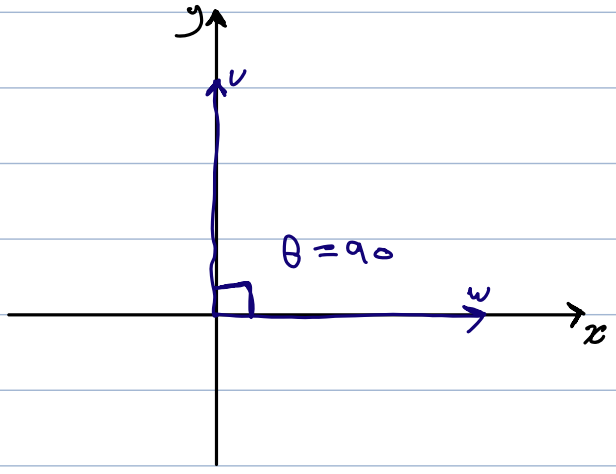
1- Positive

$v.w = 2598000$



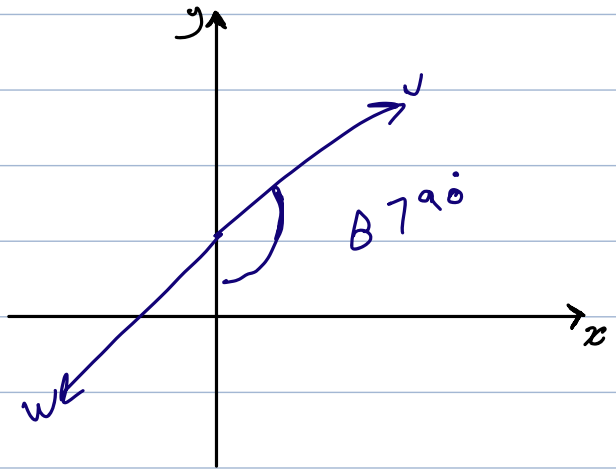
2 - Zero

$v.w = 0$



3 - Negative

$v.w < 0$



5. Create two vectors and test:
   - Are they linearly independent?
   - Do they form a basis for R^2?

$v = [2, 1]$

$w = [1, 3]$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = (2 \times 3) - (1 \times 1) = 6 - 1 = \underline{\underline{5}}$$

Since the determinant is $\underline{\underline{5}}$, the vectors are linearly independent.
They form a basis for R^2 because they are linearly independent.

6. Create a linear combination:

a * v + b * w

$a = 2$

$b = 3$

$2v = 2[2, 1] = [4, 2]$

$3w = 3[1, 3] = [3, 9]$

$a * v + b * w = [4, 2] + [3, 9] = [7, 11]$

7. Apply a linear transformation using a matrix:
   - Scaling
   - Rotation or shear

$$Av = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+0 \\ 0+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$Rv = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$